

Project Planning Worksheet

To pass this course, you'll need to create a project that matches this criteria:

"Based on your understanding of the material, you're required to build and submit an open-source project that uses NVIDIA Jetson and incorporates elements of AI (machine learning or deep learning) with GPU acceleration, along with a video demonstrating the project in action. For example, you could collect your own dataset and train a new DNN model for a specific application, add a new autonomous mode to JetBot, or create a smart home / IoT device using AI - these need not be limited only to topics covered in the course. For inspiration, see the [Jetson Community Projects](#) page - the possibilities are endless!"

To pass the certification, your project will be reviewed based on the following criteria:

- **AI** - The project uses deep learning, machine learning, and/or computer vision in a meaningful way and demonstrates a fundamental understanding of creating applications with AI. Factors include the effectiveness, technical complexity, and performance of your AI solution on Jetson.
- **Impact / Originality** - The concept of your project is novel and applies AI to solve or address challenges or issues faced by yourself or society. Also, our ideas and work are either original or derivative in a significant way.
- **Reproducibility** - Any plans, code, and resources needed for someone else to build and use the project are included in the repository and are easy to follow.
- **Presentation and Documentation** - The video effectively demonstrates and explains various aspects of the project, and there exists a clear, complete README in the repository that documents any steps needed to build/run the project, along with diagrams and images.

Follow these steps to plan out your project

Part One: Brainstorming

Write down 3-5 ideas for problems that you see in the world around you that you could create an AI to help solve. You can use [student example projects](#) or [community example projects](#) for inspiration or look back on past lessons that you enjoyed.

1. Blind people cannot understand people using fingers to represent numbers, making them unable to communicate with deaf people.
2. People can't sort trash properly, the recycle bin is filled with trash that's not-recyclable.
3. People might forget to flush after going to the bathroom, causing chaos for people behind them.
4. People without the internet can't use a translator in another country, this might put them in danger, for example, someone might rob him and he can't call the cops.
- 5.

Part Two: Details

Write down the answers to these questions for your **two favorite** ideas:

AI: How would the AI work? Technically speaking what kind of network is it and how does this network work?

Idea 1:

Hand gesture convert into voice signal:

- The camera captures live video, and the AI detects any Human hand in the picture.
- Once the hand is found, the Ai detects the shape of the finger using detectnet to tell what number the hand is representing.
- The code counts the number of extended finger tips and instantly starts an Ai read out loud system to read the number out loud through an earphone.

Idea 2:

AI recycle bin:

- A camera mounted inside the trash bin takes a photo of an item when someone puts it on top of a sorting tray.
- The image is passed through the classification AI, which scans the texture, color and lots more to identify what the object is.
- The network matches the object to a preset list such as plastic bottle, soda can, or cardboard and categorizes it. Finally, a claw catches the trash and throws it into the correct bin.

Impact: What impact would this project have? Who does it impact and in what ways?

Idea 1:

This project uses an AI object detection network and OpenCV to track human hands and count extended fingers from a live camera stream. Once a number 1 to 5 is recognized, it immediately uses a Text-to-Speech system to read the digit out loud into an earphone. This allows blind people to instantly hear and understand visual finger-counting or basic sign language numbers, breaking communication barriers in their daily lives.

Idea 2:

This project uses an AI image classification network to recognize different types of waste when an item is held over a trash bin. The network scans visual features like shape and material to identify the object and instantly triggers a motor to open the correct recycling lid automatically. This removes human error from recycling, preventing trash contamination and significantly protecting the local environment.

Part Three: Resources

Now that you have thought out the impact and technical aspects of how the AI will work, it is time to map out what resources are going to be needed to complete your project.

Docs from jetson-inference: Add your documentation or tutorial link below

<https://www.kaggle.com/datasets/nikitaandriievskyi/hand-calculator-classification-data>

Example code: Add your example code below

Using the detectnet.py script for live camera inference and the train_ssd.py script from the pytorch-ssd repository for model retraining.

Datasets: If applicable, add the dataset that you will be using below

[Finger Calculator \(Classification Data\)](https://github.com/AndriievskyiN/ObjectDetectionHandCalculator) This dataset is a part of my object detection project that you find out more about on GitHub: <https://github.com/AndriievskyiN/ObjectDetectionHandCalculator>

Miscellaneous: Add any other resources you might need for your project below.

A camera connected with my orin to provide it visual information to analyze later.

Part Four: Documentation

Video: Write down any key points that you want to add into your video below

- Show the Jetson board and the USB camera.
- Hold a plastic bottle in front of the camera.
- Show the web screen turning green for recycling.
- Show the screen turning red for bad trash.

Documentation: Write down any key points that you want to make sure are in your readme doc.

- List the hardware like Jetson and USB camera.
- Write the command to start the Docker container.
- Tell people to type `python3 web_ui.py`.
- Give the IP address link to open the webpage.

Reproducibility: How could your project be reproduced or run on another machine. Make sure to remember all steps that make your project work.

1. Download my project folder from my GitHub page.
2. Make sure you have the jetson-inference Docker ready.
3. Mount the folder into the Docker container.
4. Run the `web_ui.py` code and open your browser.

